EE4-05
EE9-CS7-22
EE9-SO7

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2017

MSc and EEE PART IV: MEng and ACGI

TRAFFIC THEORY & QUEUEING SYSTEMS

Corrected copy

Tuesday, 23 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D.P. Mandic

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Special instructions for students

1. Recursive evaluation of Erlang Loss formula:

$$E_N(\rho) = \frac{\rho E_{N-1}(\rho)}{N + \rho E_{N-1}(\rho)}$$
$$E_0(\rho) = 1$$

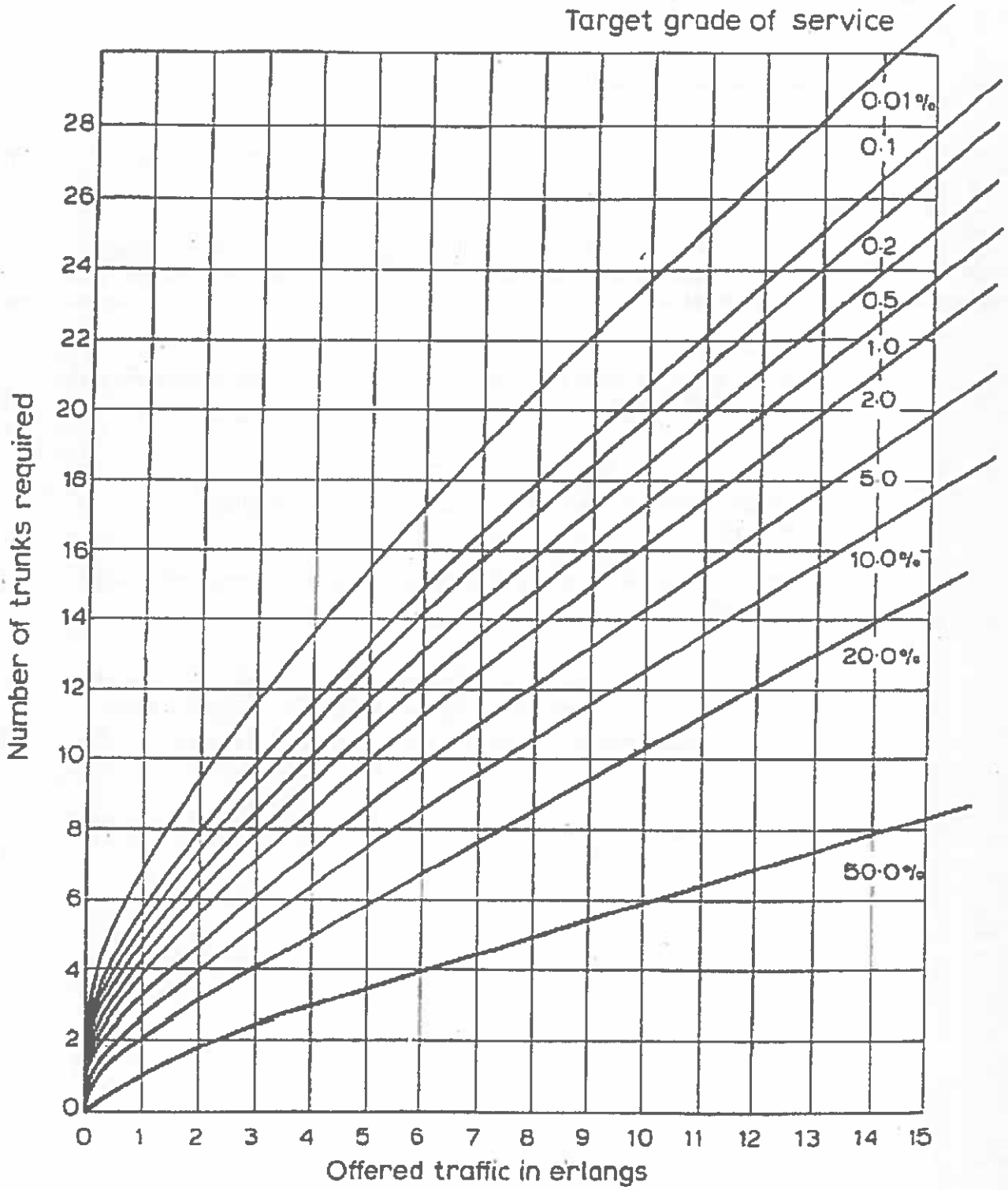
2. Recursive evaluation of Engset Loss formula (for a given M and $p = \alpha/1+\alpha$):

$$e_N = \frac{(M - N + 1)\alpha e_{N-1}}{N + (M - N + 1)\alpha e_{N-1}}$$
$$e_0 = 1$$
$$\alpha = \lambda/\mu$$

3. Traffic capacity on the basis of Erlang B formula (next page).

4. Expected residual time

$$E[R] = \frac{1}{2}\lambda E[S^2] = \frac{1}{2}\sum_{k=1}^{m}\lambda_k E[S_k^2]$$



Traffic capacity on basis of Erlang B. formula.

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The Questions

1.

a) In the context of modelling a Markov process:

i) Give two examples of Birth/Death (B/D) processes known to you. [6]

ii) Choose one of the B/D introduced in i) and derive the corresponding Global Balance Equations (GBE) and the Local Balance Equations (LBE). [6]

iii) Starting with, for example, the derived LBE in ii) show that the LBE and GBE are equivalent for the same process. [2]

b) A multichannel link is required to be designed with a loss probability of $B_c = 0.02$.

The link average calling rate is 1800 calls per hour and the average call duration is 3 minutes.

i) Using the provided graph of the system capacity based on the Erlang B formula, and for large values of the offered traffic ρ, derive a linear approximation of the number of trunks required (N) as a function of ρ. [2]

ii) Using the approximation derived in i) estimate: the size (N) of the link and the total carried traffic. [4]

2.

a) For a $M/G/1$ queue system:

i) Define, for the $i-th$ arrival, the residual waiting time variable and the mean waiting time for a $M/G/1$ queue system. [2]

ii) Show that the expected waiting time can be expressed as a function of the expected residual time. [4]

iii) Show that the expected waiting time is a function of the mean squared service time. [3]

b) For a $M/M/K$ queue system:

i) Derive the queue length distribution for delayed arrivals. [3]

ii) Derive the expected queue length for delayed arrivals. [3]

iii) Derive the variance of the queue length for delayed arrivals. [3]

iv) Derive the unconditional queue length distribution for the $M/M/K$ queue system. [2]

Note: state clearly all the steps in your derivations.

3.

a) Consider the following problem:

A Poisson stream of messages with mean rate of $\lambda = 250$ messages per second arrives to a queue with infinite buffer capacity. Sixty percent (60%) of the messages are composed of a single size 160 bits long packet. The reminder 40% of messages consists of two packets of size 160 bits long each.

The channel transmission rate is 64 Kbits/s.

For a non-pre-emptive priority treatment of messages of a single size 160 bits long packets:

i) Derive the mean transit time for the single packet messages. [6]

ii) Derive the mean transit time for two packets messages. [6]

iii) Derive the overall mean message transit time. [2]

b) Using Fig 3.1 as a reference:

Assume that the numbers of channels of Link 1 is $M = 12$ and that the number of channels of Link 2 is $N = 8$.

Assume also that the offered traffic to the system is $\rho = 10$ Erlangs.

i) Calculate the Link 1 blocking probability B_1 and, the Link 2 blocking probability B_2. [3]

ii) Derive the carried traffic on Link 1 and the carried traffic on Link 2. [3]

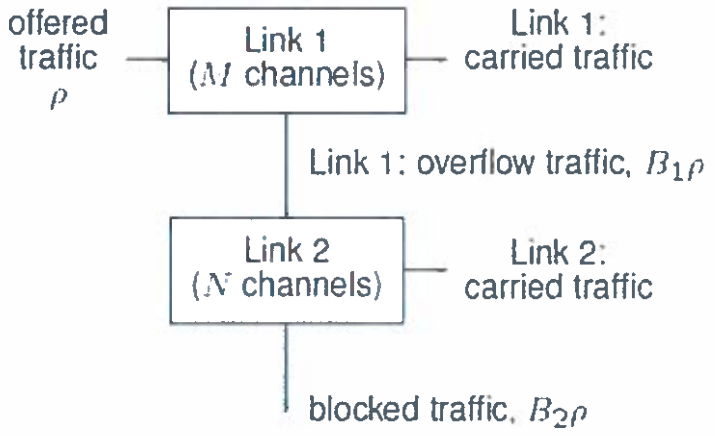


Figure 3.1.

4.

a) For an $M/M/K/N$ queuing system, the buffer has a limited capacity measured in buffer stage units. Also consider that the following characteristics of the system are known to you:

- the failure rate of each server is λ,
- the failure rate of any of the buffer stage is γ,
- the repair rate of the servers is μ,
- the repair rate of the buffer is τ.

The system is in a faulty condition if any of the buffer stage is in failure and/or all the servers units fail.

i) Define the state space of the system. [4]

ii) Construct the base model of the associated Performability model. Clearly show all the transition rates. [6]

iii) Clearly identify the faulty states. [2]

b) In the context of a fluid flow model (FFM) framework:

i) State clearly and discuss the assumption made in a FFM framework with respect to the packets being generated during an active period of an ON-OFF packet source. [2]

ii) For a statistical multiplexer's access buffer: derive step by step an equation that you can use to obtain the stationary probability that the buffer occupancy is greater than x, given that there are i ON-OFF sources active. [6]